# Provers and Solvers

## Lecture 2: Superposition

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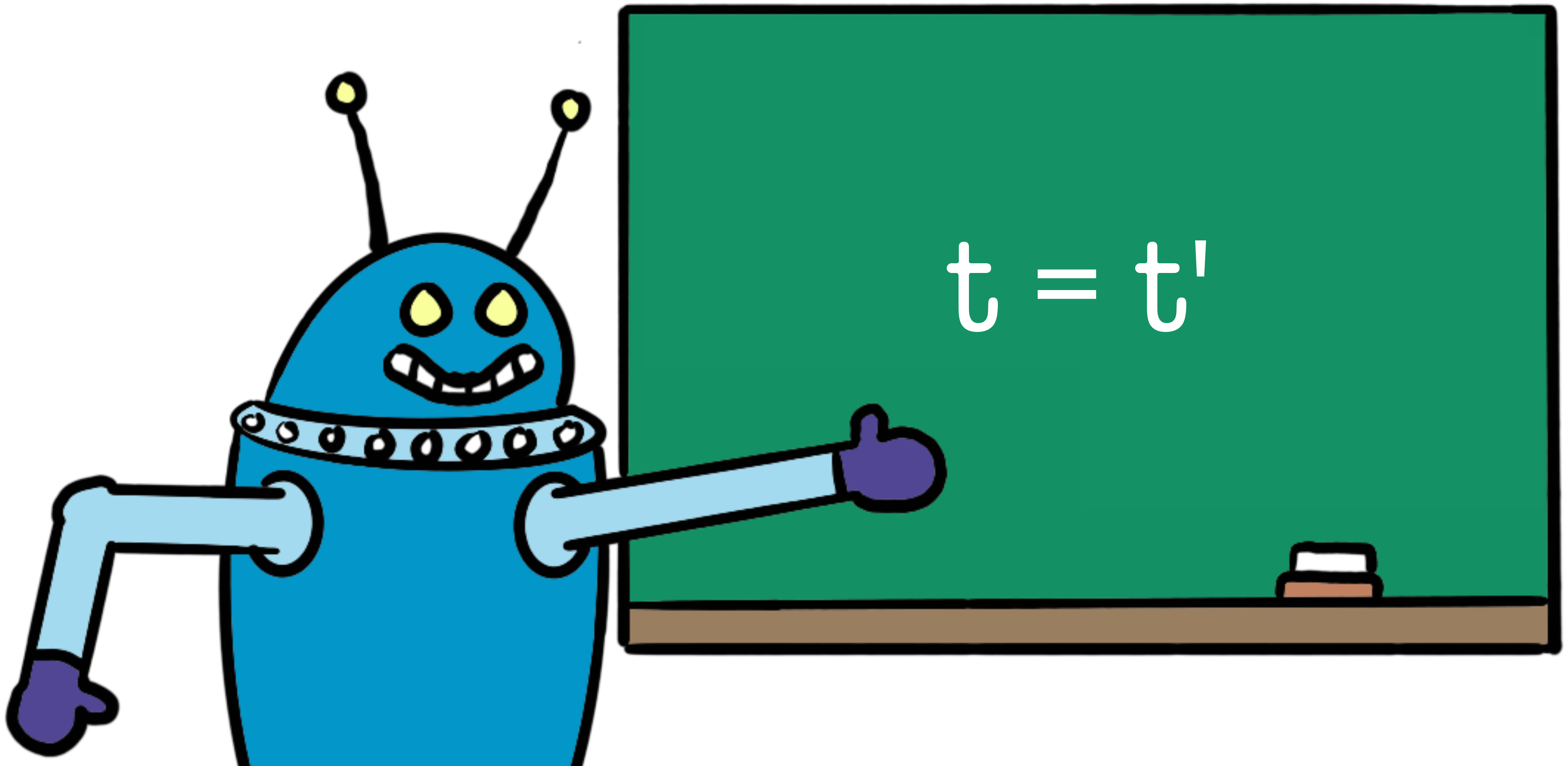
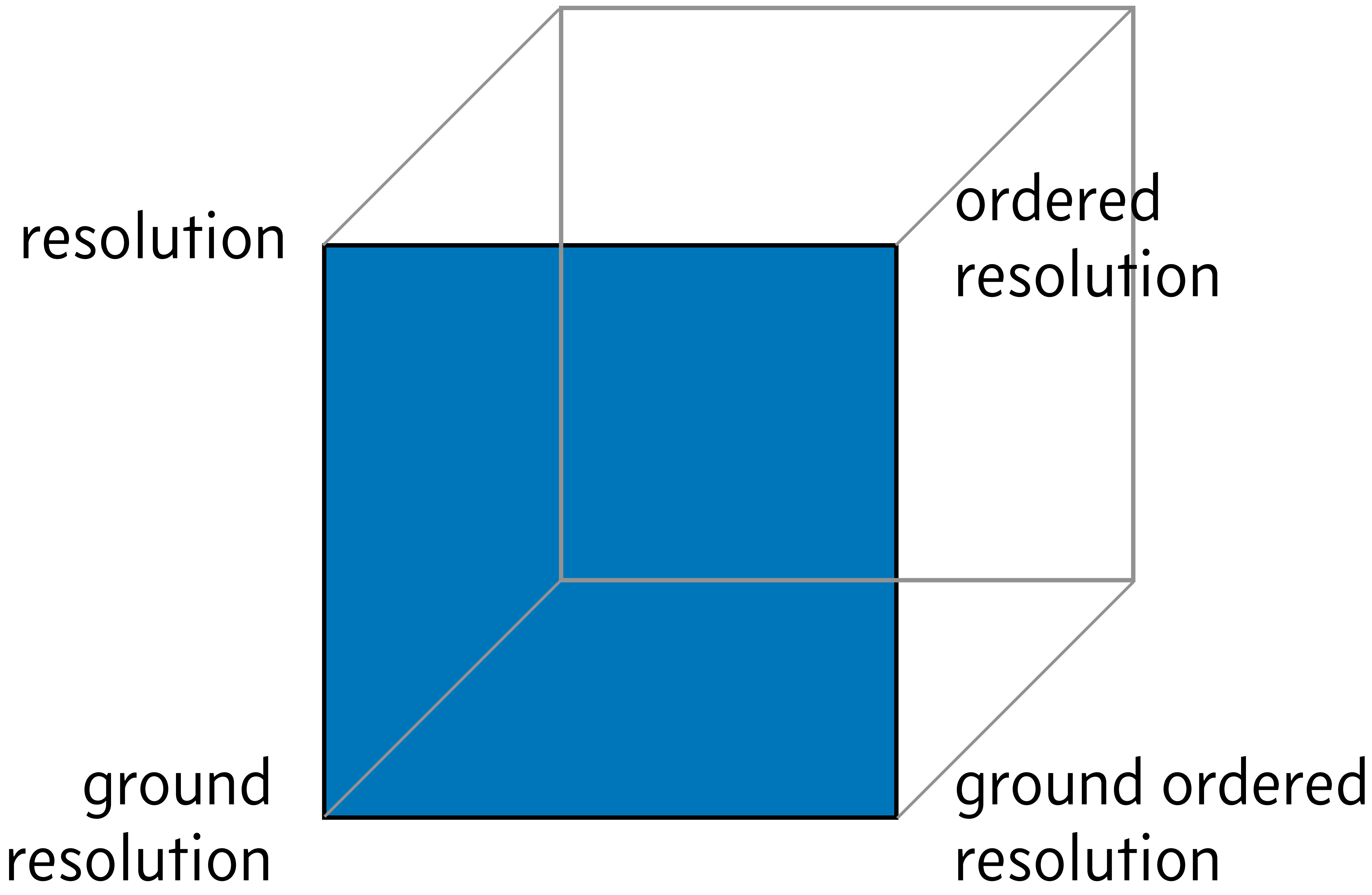
LMU Munich



# Outline of These Lectures

1. Resolution
2. **Superposition**
3. λ-Superposition
4. AVATAR

# Relations between the Calculi



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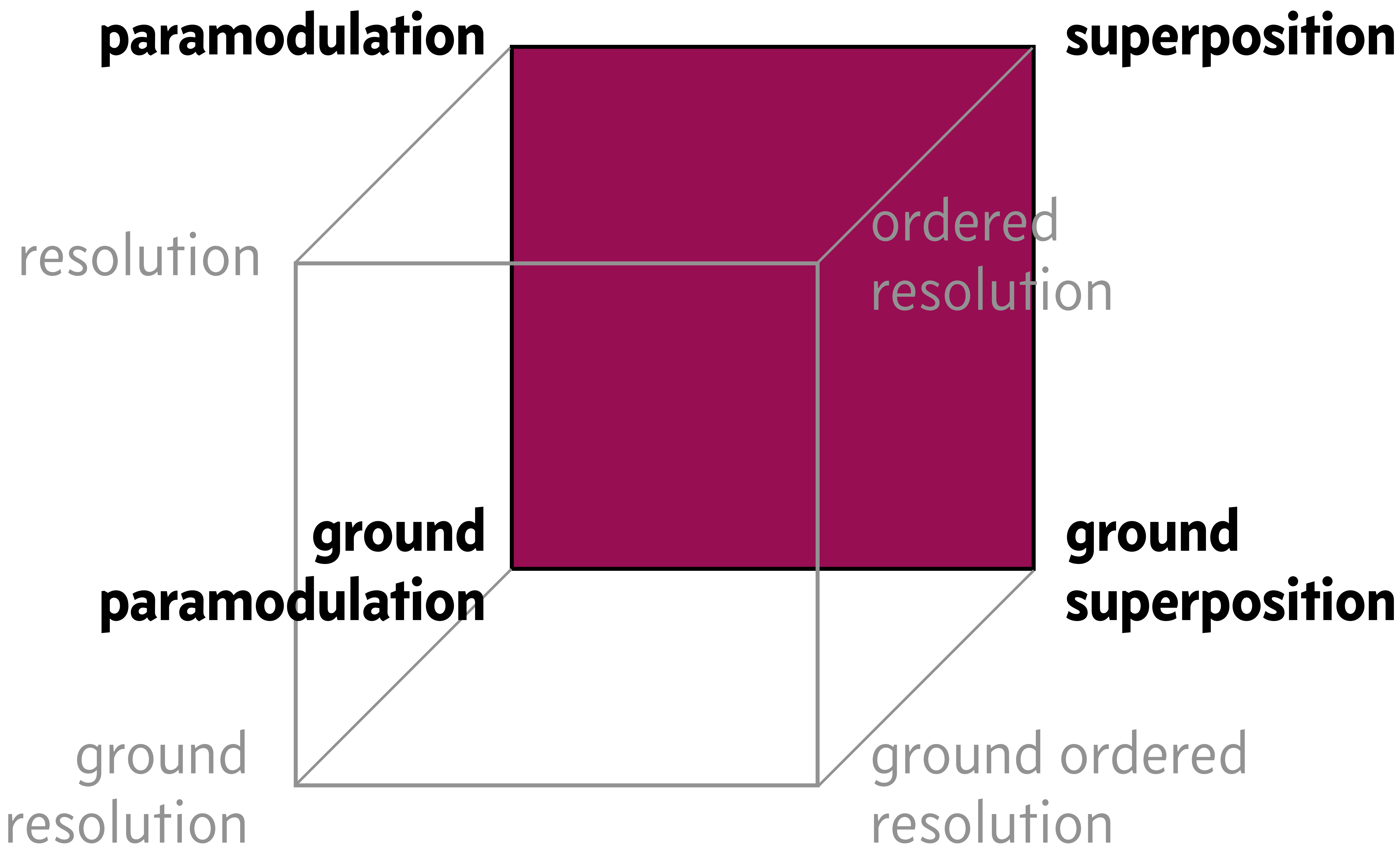

# Axiomatizing Equality

To reason about equality with resolution
we need to specify its properties:

| | |
|---|---|
| reflexivity: | $\forall x.\ x = x$ |
| symmetry: | $\forall x, y.\ x = y \Rightarrow y = x$ |
| transitivity: | $\forall x, y, z.\ (x = y \wedge y = z) \Rightarrow x = z$ |
| congruence: | $\forall x, y.\ x = y \Rightarrow \mathsf{f}(x) = \mathsf{f}(y)$ |
| | $\vdots$ |

# Alternative: Built-in Equality

Equality is treated specially by the proof calculus

Reflexivity, symmetry, transitivity, and congruence are not axiomatized

W.l.o.g.: Equality is the only predicate
$p(x)$ ➡ $p(x) = \text{true}$ $\neg p(x)$ ➡ $p(x) \neq \text{true}$

New notion of literals: $t = t'$, $t \neq t'$

# Syntax of First-Order Logic with Equality

Given a **signature**, consisting of

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The **terms** are defined by these two rules:

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- If $f$ is a function symbol and $t_1, \ldots, t_n$ are terms, then $f(t_1, \ldots, t_n)$ is a term

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The **formulas** are defined by these rules:

- $t_1$ and $t_2$ are terms, then $t_1 = t_2$ is a formula
- If $F$ and $G$ are formulas and $x$ is a variable, then $\bot$, $\top$, $\neg F$, $F \wedge G$, $F \vee G$, $F \Rightarrow G$, $\forall x.\ F$, and $\exists x.\ F$ are formulas

# Semantics of First-Order Logic with Equality

An **interpretation** $\mathcal{J}$ consists of

- a nonempty set $D$, called **domain**
- an operator ${}^{\mathcal{J}}$ for the symbols:
  - $f^{\mathcal{J}} : D^n \to D$ for every $n$-ary function symbol $f$

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The interpretation $\mathcal{J}$ is lifted to ground terms as follows:

$$[\![f(t_1,\dots,t_n)]\!]^{\mathcal{J}} = f^{\mathcal{J}}([\![t_1]\!]^{\mathcal{J}},\dots,[\![t_n]\!]^{\mathcal{J}})$$

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$$\llbracket t_1 = t_2 \rrbracket^{\mathfrak{J}} \text{ iff } \llbracket t_1 \rrbracket^{\mathfrak{J}} = \llbracket t_2 \rrbracket^{\mathfrak{J}}$$
$$\llbracket \bot \rrbracket^{\mathfrak{J}} \text{ iff false}$$
$$\llbracket \top \rrbracket^{\mathfrak{J}} \text{ iff true}$$
$$\llbracket \neg F \rrbracket^{\mathfrak{J}} \text{ iff not } \llbracket F \rrbracket^{\mathfrak{J}}$$
$$\llbracket F \wedge G \rrbracket^{\mathfrak{J}} \text{ iff } \llbracket F \rrbracket^{\mathfrak{J}} \text{ and } \llbracket G \rrbracket^{\mathfrak{J}}$$

etc.

# Syntax of Clausal First-Order Logic with Equality

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- If $L_1, \ldots, L_n$ are literals, then $L_1 \vee \cdots \vee L_n$ (viewed as a multiset) is a clause

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☞

- No $\top$, $\wedge$, $\Rightarrow$, $\forall$, $\exists$
- Variables are understood as "for all"

# Ground Paramodulation Rule for Single Literals

*Main rule:*

If the clauses

$t = t'$
$L[t]$

are contained in $\mathbb{F}$,
then **add** the clause

$L[t']$

to $\mathbb{F}$

# Ground Paramodulation Rules for Clauses

*Main rule:*

If the clauses

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*Side rule*:*

If the clause

$$C \vee t \neq t$$

is contained in $\mathbb{F}$,
then add the clause

$$C$$

to $\mathbb{F}$

* and another side rule

# Nonground Paramodulation Rules

*Main rule:*

If the clauses

$$C \lor t = t'$$
$$D \lor L[u]$$

are contained in $\mathbb{F}$
$\sigma = \mathbf{mgu}(t, u)$, and ***u* is not a variable**,
then add the clause

$$\sigma(C \lor D \lor L[t'])$$

to $\mathbb{F}$

*Side rule*:*

If the clause

$$C \lor t \neq u$$

is contained in $\mathbb{F}$
and $\sigma = \mathbf{mgu}(t, u)$,
then add the clause

$$\sigma(C)$$

to $\mathbb{F}$

* and another side rule

# Example: Nonground Paramodulation Rule

Given the clauses

$$C(y) \vee \mathrm{f}(\mathrm{a}, y) = \mathrm{c} \qquad \mathrm{g}(\mathrm{f}(x, \mathrm{b})) \neq \mathrm{d} \vee D(x)$$

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$$C(\mathrm{b}) \vee \mathrm{f}(\mathrm{a}, \mathrm{b}) = \mathrm{c} \qquad \mathrm{g}(\mathrm{f}(\mathrm{a}, \mathrm{b})) \neq \mathrm{d} \vee D(\mathrm{a})$$

# Example: Nonground Paramodulation Rule

Given the clauses

$$C(y) \lor f(a, y) = c \qquad g(f(x, b)) \neq d \lor D(x)$$

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$$C(b) \lor D(a) \lor g(c) \neq d$$

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1. Unify the f terms, yielding $\sigma = \{x \mapsto a, y \mapsto b\}$
2. Add $\sigma(C(y) \lor D(x) \lor g(c) \neq d)$ to $\mathbf{F}$

# Example: Inverse of Pi

*Claim:*

Assuming that $\pi \neq 0$ and that $x^{-1} = 1/x$ for all $x \neq 0$,
we have $|\pi^{-1}| = |1/\pi|$.

# Example: Inverse of Pi

$$\mathsf{pi} \neq \mathsf{zero} \land (\forall x.\ x \neq \mathsf{zero} \Rightarrow \mathsf{inv}(x) = \mathsf{div}(\mathsf{one}, x))$$
$$\Rightarrow \mathsf{abs}(\mathsf{inv}(\mathsf{pi})) = \mathsf{abs}(\mathsf{div}(\mathsf{one}, \mathsf{pi}))$$

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$$\mathrm{pi} \neq \mathrm{zero} \qquad x = \mathrm{zero} \vee \mathrm{div}(\mathrm{one}, x) = \mathrm{inv}(x)$$
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If the ground clause set $\mathbb{F}$ is initially unsatisfiable
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If the ground clause set $\mathbb{F}$ is initially unsatisfiable
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**Proof idea:**

- By fairness, the limit is saturated
  (all inferences have been done or are redundant)
- By static completeness,
  the limit contains $\bot$

# Static Completeness of Ground Paramodulation

If a clause set $\mathtt{F}$ is unsatisfiable and saturated,
then $\mathtt{F}$ contains $\perp$

# Static Completeness of Ground Paramodulation

If a clause set $\mathbb{F}$ is unsatisfiable and saturated,
then $\mathbb{F}$ contains $\bot$

**Contrapositive:** If a clause set $\mathbb{F}$ is saturated
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# Static Completeness of Ground Paramodulation

If a clause set $\mathbb{F}$ is unsatisfiable and saturated, then $\mathbb{F}$ contains $\perp$

**Contrapositive:** If a clause set $\mathbb{F}$ is saturated and does not contain $\perp$, then $\mathbb{F}$ has a model

**Proof idea:**
We build an **equality term model** incrementally based on the clause order

# What Is an Equality Term Model?

An equality term model is a model in which all terms are interpreted as their equivalence classes:

$$[\![f(t_1,\ldots,t_n)]\!]^{\mathfrak{I}} = [f(t_1,\ldots,t_n)]_R$$

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Such models can be represented by the
equations that specify the relation $R$

E.g. $R = \{\mathrm{f(a)} = \mathrm{a}\}$ ensures $[\mathrm{f(a)}]_R = [\mathrm{a}]_R$ but also
$[\mathrm{f(f(a))}]_R = [\mathrm{a}]_R$, $[\mathrm{g(f(a))}]_R = [\mathrm{g(a)}]_R$, etc.

# Static Completeness of Ground Paramodulation

**Contrapositive:** If a clause set $\mathbf{F}$ is saturated and does not contain $\bot$, then $\mathbf{F}$ has a model

$\mathbf{F}$

$g = a \lor h = b$

$f(g) \neq f(a) \lor k \neq c$

$f(g) \neq f(a) \lor k = c$

$g \neq a \lor h = b$

$g = a$

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# Ground Superposition Rules

*Main rule:*

If the clauses

$$C \lor \boxed{t} = t'$$
$$D \lor \boxed{s[t]} \left\{ \begin{smallmatrix} = \\ \neq \end{smallmatrix} \right\} s'$$

are contained in $\mathbb{F}$ and
$t$ and $s[t]$ are **maximal side of maximal literal**,
then add the clause

$$C \lor D \lor s[t'] \left\{ \begin{smallmatrix} = \\ \neq \end{smallmatrix} \right\} s'$$

to $\mathbb{F}$

*Side rule*:*

If the clause

$$C \lor t \neq t$$

is contained in $\mathbb{F}$
and $t \neq t$ is **maximal**,
then add the clause

$$C$$

to $\mathbb{F}$

* and another side rule

# Ground Redundancy Criterion

A ground clause $C$ is redundant w.r.t. ground $\mathtt{F}$ if $C_1, \ldots, C_n \in \mathtt{F}$ are all smaller than $C$ and they together entail $C$

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**Examples:**

- b = a makes f(b) = f(a) redundant
- $\boxed{\text{b}}$ = a and a = $\boxed{\text{c}}$ make b = c redundant

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- $\boxed{\text{b}}$ = a and a = $\boxed{\text{c}}$ make b = c redundant

The completeness proof still goes through

# Examples of Redundancy Rules

*Tautology deletion:*

If a clause of either form

$$C \lor t = t$$
$$C \lor t = u \lor t \neq u$$

is contained in $\mathtt{F}$,
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*Literal rewriting:*

If the two clauses

$$t = t'$$
$$D \lor L[t]$$

are contained in $\mathbb{F}$,
$t > t'$, and another condition,
then **replace** $D \lor L[t]$ with $D \lor L[t']$ in $\mathbb{F}$

# Nonground Superposition Rules

Not given here:
Combine the **unification** and **variable condition** of nonground paramodulation with the **order restrictions** of ground superposition

# Nonground Redundancy Criterion

A nonground clause $C$ is redundant w.r.t. nonground $\mathtt{F}$
if each clause in ground($C$) is redundant w.r.t. ground($\mathtt{F}$)

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A nonground clause $C$ is redundant w.r.t. nonground $\mathtt{F}$
if each clause in ground($C$) is redundant w.r.t. ground($\mathtt{F}$)

**Examples:**

- $\mathrm{b}(x) = \mathrm{a}(x)$ makes $\mathrm{f}(\mathrm{b}(x)) = \mathrm{f}(\mathrm{a}(x))$ redundant
- $\boxed{\mathrm{b}(x)} = \mathrm{a}(x)$ and $\mathrm{a}(x) = \boxed{\mathrm{c}(x)}$ make $\mathrm{b}(x) = \mathrm{c}(x)$ redundant

# Soundness of Nonground Superposition

If ground($C \vee A$) and ground($\neg A' \vee D$) are all true,
then each clause in ground($\sigma(C \vee D)$) is true

**Proof idea:** Similar to the ground case

# Completeness of Nonground Superposition

If the nonground clause set $\mathtt{F}$ is initially unsatisfiable
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then $\mathtt{F}$ will eventually contain $\bot$

# Completeness of Nonground Superposition

If the nonground clause set $\mathbf{F}$ is initially unsatisfiable
and inferences are performed fairly,
then $\mathbf{F}$ will eventually contain $\bot$

**Proof idea:**

- By fairness, the limit is saturated
  (all inferences have been done or are redundant)
- By static completeness,
  the limit contains $\bot$

# Static Completeness of Nonground Superposition

If a clause set $\mathbb{F}$ is unsatisfiable and saturated,
then $\mathbb{F}$ contains $\perp$

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**Proof idea:**

- We show that ground($\mathbb{F}$) is saturated
  (nonground inferences are overapproximation)
- By static completeness of ground superposition,
  ground($\mathbb{F}$) contains $\perp$
- Hence $\mathbb{F}$ contains $\perp$

# Static Completeness of Nonground Superposition

If a clause set $\mathbf{F}$ is unsatisfiable and saturated,
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- We show that ground($\mathbf{F}$) is saturated (nonground inferences are overapproximation)
- By static completeness of ground superposition, ground($\mathbf{F}$) contains $\bot$
- Hence $\mathbf{F}$ contains $\bot$

**Twist:** inferences below variables

# Approximation of Ground Inferences

$\mathbb{F}$

$\mathsf{g}(x) = \mathsf{f}(x)$

$\mathsf{g}(y) \neq \mathsf{d}$

saturated

# Approximation of Ground Inferences

$\mathtt{F}$

$g(x) = f(x)$

$g(y) \neq d$

saturated

$\mathrm{ground}(\mathtt{F})$

$g(a) = f(a)$
$g(b) = f(b)$
$g(c) = f(c)$
$\vdots$

$g(a) \neq d$
$g(b) \neq d$
$g(c) \neq d$
$\vdots$

# Approximation of Ground Inferences

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saturated?

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$f(b) \neq d$?

saturated?

# Approximation of Ground Inferences

$\mathtt{F}$

$g(x) = f(x)$ $\qquad$ $g(y) \neq d$

$f(x) \neq d$

saturated

$\mathrm{ground}(\mathtt{F})$

$g(a) = f(a)$ $\qquad$ $g(a) \neq d$

$g(b) = f(b)$ $\qquad$ $g(b) \neq d$

$g(c) = f(c)$ $\qquad$ $g(c) \neq d$

$\vdots$ $\qquad$ $\vdots$

$f(b) \neq d$?

saturated?

# Ground Inferences below Variables

$\mathtt{F}$

$b = a$ $\qquad$ $f(x) \neq d$

saturated

# Ground Inferences below Variables

| $\mathtt{F}$ | | $\text{ground}(\mathtt{F})$ | |
|---|---|---|---|
| $b = a$ | $f(x) \neq d$ | $b = a$ | $f(a) \neq d$ |
| | | | $f(b) \neq d$ |
| | | | $f(c) \neq d$ |
| | | | $\vdots$ |
| saturated | | | |

# Ground Inferences below Variables

$\mathtt{F}$

$b = a$

$f(x) \neq d$

saturated

ground($\mathtt{F}$)

$b = a$

$f(a) \neq d$
$f(b) \neq d$
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$\vdots$

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$\mathtt{F}$

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???

saturated

$\mathrm{ground}(\mathtt{F})$

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$f(b) \neq d$
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$\vdots$

$f(a) \neq d$?

saturated?

# Ground Inferences below Variables

**F**

$b = a$

$f(x) \neq d$

???

saturated

ground(**F**)

$b = a$

$f(a) \neq d$

$f(b) \neq d$

$f(c) \neq d$

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saturated?

# Ground Inferences below Variables

$\mathtt{F}$

$b = a$ $\quad$ $f(x) \neq d(x)$

saturated

# Ground Inferences below Variables

$F$

$b = a$

$f(x) \neq d(x)$

saturated

ground($F$)

$b = a$

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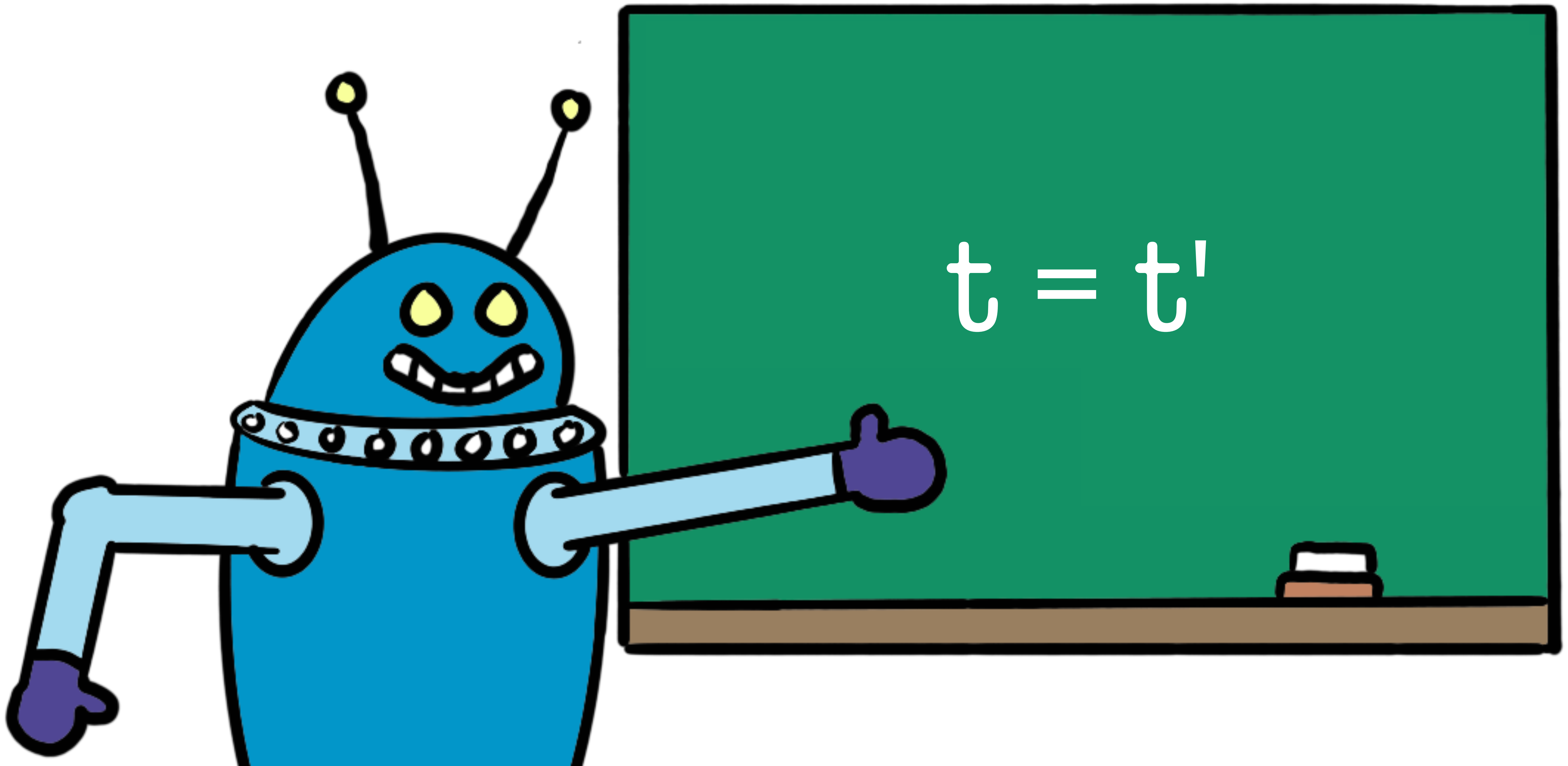
# Provers and Solvers

## Lecture 2: Superposition

**Jasmin Blanchette**
LMU Munich



# Blah